

15-Nearest Neighbor Classifier

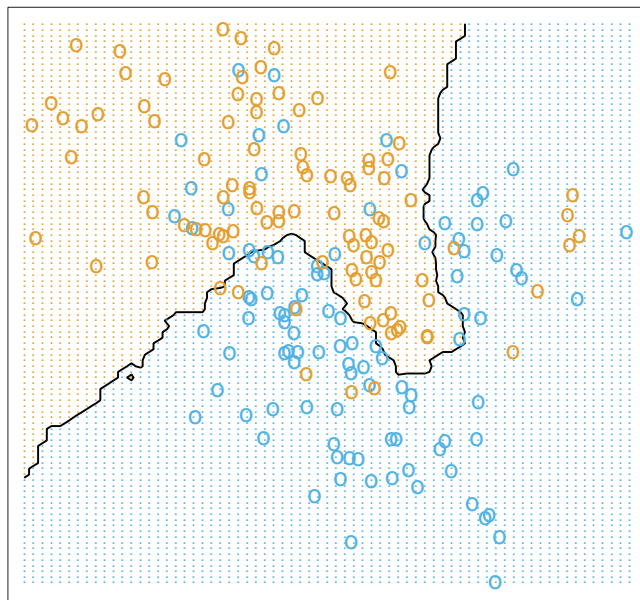


FIGURE 2.2. The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable (BLUE = 0, ORANGE = 1) and then fit by 15-nearest-neighbor averaging as in (2.8). The predicted class is hence chosen by majority vote amongst the 15-nearest neighbors.

In Figure 2.2 we see that far fewer training observations are misclassified than in Figure 2.1. This should not give us too much comfort, though, since in Figure 2.3 *none* of the training data are misclassified. A little thought suggests that for k -nearest-neighbor fits, the error on the training data should be approximately an increasing function of k , and will always be 0 for $k = 1$. An independent test set would give us a more satisfactory means for comparing the different methods.

It appears that k -nearest-neighbor fits have a single parameter, the number of neighbors k , compared to the p parameters in least-squares fits. Although this is the case, we will see that the *effective* number of parameters of k -nearest neighbors is N/k and is generally bigger than p , and decreases with increasing k . To get an idea of why, note that if the neighborhoods were nonoverlapping, there would be N/k neighborhoods and we would fit one parameter (a mean) in each neighborhood.

It is also clear that we cannot use sum-of-squared errors on the training set as a criterion for picking k , since we would always pick $k = 1$! It would seem that k -nearest-neighbor methods would be more appropriate for the mixture Scenario 2 described above, while for Gaussian data the decision boundaries of k -nearest neighbors would be unnecessarily noisy.